

Jisoo Jang

Ph.D. Candidate

Graduate School of Data Science
Seoul National University
☎ (+82) 1098303450
✉ simonjisu@gmail.com
📄 simonjisu.github.io
🌐 [simonjisu](#)
in [soopace](#)

Education

- 2022–2026 **Doctor of Philosophy in Data Science**, Graduate School of Data Science, Seoul National University (Expected Graduation: 2026.08), Seoul.
Large Language Models (LLMs) and Context Engineering, LLM-Driven Natural Language Interfaces for Databases, Natural Language to Structured Query Language (NL2SQL)
- 2020–2021 : **Master in Data Science**, Graduate School of Data Science, Seoul National University, Seoul.
- 2010–2018 : **BA in Economics & Finance**, College of Economics and Finance, Hanyang University, Seoul.

Publications

In Conference Proceedings

- 2026 Joohyun Lee, Ghita Benboubker, JungKwan Han, Jisoo Jang, and Wen-Syan Li. Pathfinder: Context engineering and knowledge management for domain-specific horizontal reasoning. In *42nd IEEE International Conference on Data Engineering (Demo Track)*, pages 0–0, 2026.
- 2025 Jisoo Jang and Wen-Syan Li. Retrieval-augmented nl2sql generation with data-centric query capsules for enterprise applications. In *Proceedings of the 2025 Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region*, pages 123–132, 2025.
- 2024 Jisoo Jang and Wen-Syan Li. Au-rag: agent-based universal retrieval augmented generation. In *Proceedings of the 2024 Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region*, pages 2–11, 2024.
- 2023 Jisoo Jang, Mina Kim, Tien-Cuong Bui, and Wen-Syan Li. Toward interpretable machine learning: Constructing polynomial models based on feature interaction trees. In *Pacific-Asia Conference on Knowledge Discovery and Data Mining*, pages 159–170. Springer, 2023.

In Conference Proceedings(Under Review)

- 2026 Jisoo Jang, Mingyoo Son, Jungkwan Han, and Wen-Syan Li, Strategy Map: Milestone-Guided Process State for Dependency-Sensitive Tool-Use Agents, In NeurIPS 2026.
- 2026 Jisoo Jang, Mingyoo Son, and Wen-Syan Li, When Should a Routing Decision Be Re-examined? A Diagnostic Study of Model Switching During Reasoning, In NeurIPS 2026.
- 2026 Jisoo Jang, Mingyoo Son, and Wen-Syan Li, Trace2Map: Milestone-Guided Context Management for Tool-Use Agent, In CIKM 2026.

Journal Article(Under Review)

- 2026 Jisoo Jang, Mingyoo Son, Ghita Benboubker, and Wen-Syan Li, NL2SQL+: An Extended Framework for SQL Generation with On-Demand Embedded ML Functions, IEEE Access.

arXiv

- 2026 Jisoo Jang Wen-Syan Li. Twinbi: An agentic digital twin for efficient augmented interactions with business intelligence dashboards. *arXiv preprint arXiv:2606.13731*, 2026.
- 2025 Jisoo Jang, Tien-Cuong Bui, Yunjun Choi, and Wen-Syan Li. Prompt tuning for natural language to sql with embedding fine-tuning and rag. *arXiv preprint arXiv:2511.08245*, 2025.

Projects

2024.11 – **Financial Q&A Chatbot Based on Financial Reports.**

- 2025.01 (3M)
- Building a financial Q&A chatbot based on financial reports in collaboration between SNU KDT and Naver Cloud
 - Led a team by helping define the problem, design experiments, and manage the project through weekly meetings
 - Explained concepts of LLM and RAG, and guided system architecture design by applying the LLM Compiler paper

2021.04 – **Metric for Deep Learning Visualization Technique.**

- 2021.06 (2M)
- Proposed l-metric, a metric to evaluate neuron visualization as part of Explainable AI techniques
 - Conducted experiments to verify whether an additional tool (grid-spec activation camera) improves evaluation
 - Achieved a 28% improvement over baseline visualization when using the additional tool — [paper link](#)
 - Role: Implemented the grid-spec activation camera

2020.11 – **Improving Cleanness of Dishwasher.**

- 2021.11 (12M)
- A project with SK Networks to improve dishwasher cleaning performance using pattern clustering on turbidity sensor data

2020.10 – **Battery Life Expectation.**

- 2021.02 (5M)
- Predicted the remaining battery life of a logistics container tracking device by solving it as a regression problem for voltage values

Awards

2025.10 SIGIR Student Travel Grant Award

2024.10 SIGIR Student Travel Grant Award

2023.08 The 11th Ministry of Trade, Industry, and Energy Public Data Utilization BI Contest - 3rd prize in the “Public Domain” Area

Lectures

2024.08 A Guide to Building LLM-Based Applications: Prompt Engineering and RAG.
@ Jeonju IT & CT Industry Promotion Agency

2022.07 Introduction to DeepLearning with PyTorch.
@ University of Seoul

2021.07 Python & PyTorch Basic.
@ HanKyong University

2019.12 Introduction to DeepLearning with PyTorch.
@ Multicampus

Teaching Assistantship

Fall, 2022 – M3239.006700: Project for Data Science, GSDS, SNU.
Present

Fall, 2023 – M3239.002300: Prompt Engineering: Art and Science for Interactions with LLM, GSDS, SNU.
Spring, 2025

Fall, 2022 – M3239.005100: Big Data and Knowledge Management Systems 2, GSDS, SNU.
Spring, 2023

Publication (Book)

2019.06 [PyTorch for Deep Learning](#)

Work Experience

2018.02 - **Boostcourse AI**, Naver Connect, Seoul.

- 2019.08 (18M)
- Publishing AI-related content on [edwith \(changed to boostcourse\)](#) platform to make it easier for learners to study
 - Professor KyungHyun Cho's course: [Natural Language Processing with Deep Learning](#)
 - Professor Andrew Ng's Deep Learning series: [1](#), [2](#), [3](#), [4](#)
 - Planning and managing project-based learning content
 - Assisted in planning and managing the course [Deep Learning for Everyone Season 2](#)
 - Planned, set up the learning environment, and reviewed code for "Introduction to Deep Learning with [TensorFlow](#) and [PyTorch](#)"

Position of Responsibility

2025 Volunteer for conference, SIGIR-AP 2025.

2024 Student Helper to manage the in-person part, PAKDD 2024 Robust Machine Learning in Open Environments Workshop.

Interests

Attendances for Learning New Knowledge

- 2025.12 Attend the 3rd International ACM SIGIR Conference on Information Retrieval in the Asia Pacific, Xi'an, China
- 2025.08 Attend [Agentic AI Summit 2025](#) in UC Berkeley, California, USA
- 2024.12 Attend the 2nd International ACM SIGIR Conference on Information Retrieval in the Asia Pacific, Tokyo, Japan
- 2024.05 Attend the 28th Pacific-Asia Conference on Knowledge Discovery and Data Mining, Taipei, Taiwan
- 2023.05 Attend the 27th Pacific-Asia Conference on Knowledge Discovery and Data Mining, Osaka, Japan
- 2022.12 Attend 2022 IEEE International Conference on Big Data, Tokyo, Japan
- 2018.08 Attend Pycon-Korea 2018, Seoul, Korea
- 2018.01 Participate "Deep Learning Basic with Pytorch" Class @ FastCampus, Seoul, Korea
- 2017.10 Participate "Text Mining using Python" Class @ FastCampus, Seoul, Korea
- 2017.08 Attend Pycon-Korea 2017, Seoul, Korea
- 2017.05 Learned ML @ FastCampus Data Science School, Seoul, Korea

Participate or Running A Study Group

- 2022.11 (3M) Head first Rust! @ Modulabs
- 2020.05 (3M) Participate Beyond BERT @ Modulabs: reviewed [Fast-BERT](#)
- 2018.08 (9M) Ran a paper study with Yonsei University Master course students (Learning NLP) [Drive Link](#)